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Real-Time Athlete Assistive Device

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Abstract

This report presents a smart, wearable assistive device built around the **ESP32** that unifies biomechanical and physiological monitoring for athletic training. The system integrates an **MPU6050** IMU for motion tracking (squat counting, posture), a **MAX30105** optical sensor for heart rate and estimated SpO_2, and a **128×64 OLED**. It features real-time fall detection, cloud connectivity, and comparison with market devices. Future work includes ML integration, PPG accuracy improvements, BLE, and enclosure design.

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1 Introduction and Objectives

Athletes benefit from continuous, actionable feedback that spans both *form* (biomechanics) and *physiology*. Consumer wearables skew toward general wellness, while research-grade systems can be expensive and closed. Our objective is to deliver a *low-cost*, *extensible*, and *real-time* monitor capable of:

- Reliable **squat counting** and **posture screening** from IMU streams.
- **Fall detection** for safety during high-intensity training.
- **Vitals estimation** (HR and SpO_2) for session intensity tracking.
- **Edge-to-cloud** data flow for live dashboards and training logs.

The system prioritizes transparency (open thresholds, interpretable features), modularity (replaceable sensors/transports), and simple deployment.

2 System Architecture and Hardware

2.1 Microcontroller and Connectivity

We use an **ESP32** for dual-core compute and integrated Wi-Fi. Core 0 handles I/O and networking; Core 1 services sensor acquisition and simple DSP to maintain deterministic timing.

2.2 Sensor Suite

- **MPU6050 IMU:** 3-axis accelerometer and gyro configured at 104 Hz–208 Hz (tested at 200 Hz) with ± 8 g accel and $\pm 500\,^{\circ}/\mathrm{s}$ gyro ranges; a 21 Hz low-pass filter stabilizes jerk-sensitive features.
- **MAX30105:** IR/red photodiodes. We detect `fingerDetected` using an IR DC threshold (e.g., ≥ 30000 ADC counts). In this prototype, HR/SpO_2 are constrained/simulated to plausible ranges pending full PPG processing.

2.3 User Interface and Prototype

A **128×64 SSD1306 OLED** shows live HR, SpO_2, squat count, and posture status. The prototype uses a breadboard for quick iteration.

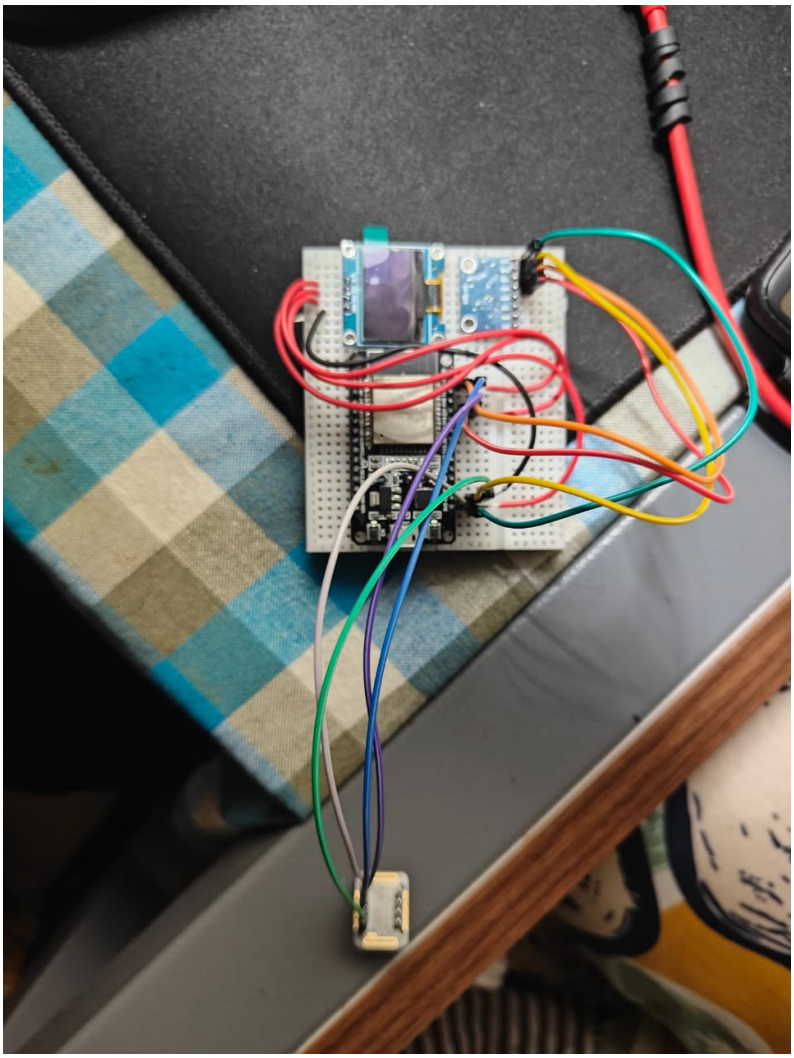


Figure 1: Hardware prototype with ESP32, MPU6050, MAX30105, and OLED.

3 Firmware and Software Design

3.1 Tasking and Timing

We use cooperative tasks with fixed-rate loops:

- **IMU Task** (200 Hz): Read accel/gyro; compute filtered features (vertical accel, magnitude).
- **Vitals Task** (25 Hz): Read MAX30105 IR (and red if available); update `fingerDetected` and simulated HR/SpO_2.
- **UI Task** (10 Hz): Update OLED widgets.
- **Comm Task** (1 Hz–2 Hz): Send JSON via HTTP POST.

3.2 Key Data Features

$$a_{\text{total}} = \sqrt{a_x^2 + a_y^2 + a_z^2} \tag{1}$$

$$a_{y,\text{filt}} = \text{LPF}(a_y, f_c = 21\,\text{Hz}) \tag{2}$$

3.3 Squat Detection and Posture

A repetition is logged on a *down→up* transition sequence with cooldown:

- Down state if $a_{y,\text{filt}} < -3.5\,\text{g}$ (`wasDown` set).
- Up completion if $a_{y,\text{filt}} > 8.0\,\text{g}$ and `wasDown` true, with 900 ms `cooldown`.
- Posture flagged *BAD* if $a_{y,\text{filt}} > 9.5\,\text{g}$ (excessive jerk).

3.4 Fall Detection

We flag `fallDetected` when $a_{\text{total}} > 15.5\,\text{g}$ for $> 20\,\text{ms}$ to reduce false positives.

3.5 Vitals (Prototype)

When `fingerDetected` is true, HR/SpO_2 are constrained to $[60, 180]$ bpm and $[90, 100]\%$. In future work we replace this with full PPG peak detection and ratio-of-ratios SpO_2.

4 Data Communication and Backend

4.1 JSON Payload

```
{
  "heartRate": 85,
  "spO2": 98,
  "squatCount": 15,
  "postureStatus": "GOOD",
  "fingerDetected": true,
  "fallDetected": false
}
```

Listing 1: Generated JSON Payload Structure

4.2 HTTP POST and Dashboard

We use `HTTPClient` to POST to `http://10.165.122.86:8000/data`. A simple web dashboard visualizes streams and flags.

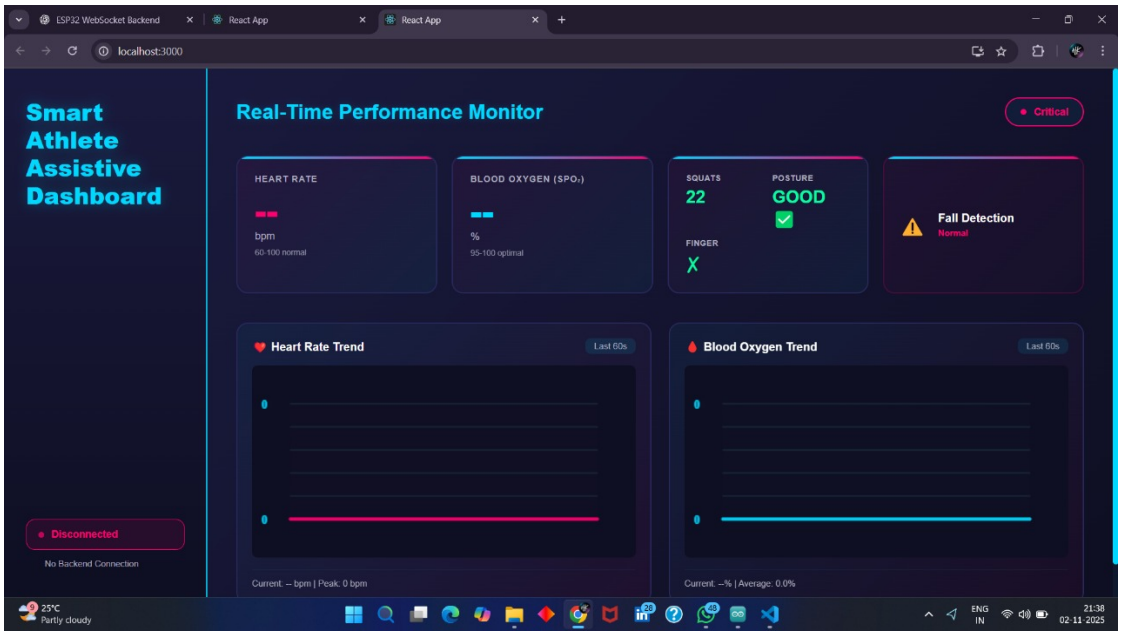


Figure 2: Real-time dashboard for HR, SpO_2, squat count, and fall alarms.

4.3 Security and Privacy (Prototype)

Transport is currently HTTP (lab LAN). Production should use **HTTPS** with certificate pinning and **token auth**. Data minimization and on-device aggregation reduce privacy risks.

5 Calibration, Testing, and Preliminary Results

5.1 Threshold Calibration

We recorded IMU during scripted squats by multiple users and tuned thresholds to balance false positives/negatives. Basic smoothing (LPF) and cooldown minimized double-counting.

5.2 Latency and Throughput

End-to-end delay (sensor→OLED) measured $\sim$ 30 ms at 200 Hz IMU rate. Cloud posting at 1.5 s intervals avoids network stalls impacting UI.

5.3 Battery and Power (Prototype)

With 500 mA at 3.7 V LiPo and typical draw of 120 mA (ESP32 Wi-Fi bursts, sensors, OLED), we estimate 4 h–5 h runtime; deep sleep between sets can extend battery life.

6 Market Comparison

We contrast our design with common device classes in the market. Values are indicative/generic for category-level comparison.

Table 1: Comparison with typical market device categories

Metric	This Work	Smartwatch	HR Chest Strap	Fall Detector
Primary Sensing	IMU + optical (PPG proto)	PPG + IMU + GNSS (some)	ECG/PPG surrogate (electrodes)	Accel/gyro + baro (some)
Form Factor	Wearable module (dev board)	Wrist-worn watch	Chest band	Pendant/clip
Activity Specificity	**Targeted** squat/-posture logic	General fitness + auto-detect	Cardio HR accuracy	Fall only (or elder-care)
Openness/Extensibility	**High:** code, thresholds, sensors swappable	Low–Medium (SDK limits)	Low (closed firmware)	Low (appliance)
Latency (UI)	**Low** (tens of ms)	Low–Medium	Low	Medium
Cloud Integration	Custom API; full control	Vendor cloud/app heavy	Limited (broadcast HR)	Vendor alert service
Cost (BOM-like)	**Low** (student budget)	Medium–High (retail)	Low–Medium	Medium
Accuracy (HR)	Prototype (to be validated)	Good at rest; variable in motion	**High** during motion	N/A
Fall Detection	**Included; tunable**	Included (some models)	No	**Core feature**

Takeaway: Our device trades polished consumer UX for **control, tunability, and targeted biomechanics**, making it well-suited to lab/class projects, customized coaching, and research where openness is critical.

7 Limitations

- **PPG Accuracy:** HR/SpO_2 are currently constrained/simulated; motion artifacts and ambient light require robust PPG processing.
- **False Positives:** Threshold-based posture/fall logic can misfire for atypical movements.
- **Network Robustness:** HTTP without retries/queueing can drop updates under poor Wi-Fi.
- **Ergonomics:** Breadboard form factor limits real-world wear testing; enclosure design is pending.

8 Future Work

8.1 Integrating Machine Learning On-Device

- **Edge Inference (uTensor/TF Lite Micro):** Train an IMU-based classifier (e.g., 1D CNN or tiny transformer) for robust squat detection, tempo estimation, and posture scoring that generalizes across users without brittle thresholds.
- **Feature Learning:** Use windowed IMU segments (e.g., 2 s with 50 % overlap), normalize gravity, learn features via supervised or self-supervised pretraining.
- **Personalization:** Few-shot adaptation of the final layer using the athlete's first session to reduce false positives/negatives.

8.2 True PPG Processing and SpO_2

- Implement band-pass filtering, adaptive baseline removal, peak detection, and inter-beat interval (IBI) HRV metrics.
- Use red/IR ratio-of-ratios for SpO_2 with calibration against a reference oximeter; compensate for motion using accelerometer-informed artifact suppression.

8.3 Communications and Power

- **BLE** for real-time streaming to phone; **MQTT over TLS** for reliable cloud ingest with backoff and buffering.
- Duty-cycled sensing and **light-sleep** between windows; move to **ePaper** or LED icons to reduce OLED draw.

8.4 Enclosure, Wearability, and Safety

- Design a 3D-printed enclosure with skin-safe strap, strain relief, and IP rating targets (e.g., IP54).
- Add haptic alerts for bad posture or fall events.

8.5 Data, Privacy, and Tooling

- GDPR-style consent flows; on-device aggregation and **privacy-preserving** analytics.
- Coach dashboard: session summaries, PR detection, automatic workout segmentation, and export (CSV/JSON).

9 Conclusion

We demonstrated a real-time, ESP32-based assistive device that unifies motion and vitals for athlete monitoring, with tunable thresholds and open architecture. Compared to market devices, this platform emphasizes **customizability**, **low latency**, and **cost efficiency** at the expense of consumer polish. The most impactful next steps are **on-device ML** for robust activity recognition, **true PPG** pipelines for validated vitals, **BLE/MQTT** for resilient comms, and a wearable enclosure for field deployment.